

DepoIndex

AI-Powered Deposition Topic Index

The Problem

- Long transcripts move between many subjects
- Discussions return to earlier topics
- Brief digressions interrupt the main thread
- Manual topic indexing is slow and inconsistent

The Solution

- Identifies meaningful topics automatically
- Determines exact start / end boundaries
- Preserves exact page:line provenance
- Validates every reference deterministically
- Produces a searchable, attorney-facing index

2,027

transcript records



51

provenance-preserving chunks



57

validated topics

Technical Architecture

End-to-end pipeline from raw PDF to a validated, searchable index



1

Canonical Extraction

- 2,027 transcript records
- Exact printed page + line kept
- Source range 7:11 → 88:13

2

Chunking

- 51 sequential chunks
- 40 transcript records / chunk
- 5 chunks batched per LLM call

3

LLM Processing

- Structured topic extraction
- Exact page:line refs required
- Output treated as non-deterministic

4

Deterministic Validation

- Every reference must exist
- start_ref must precede end_ref
- Evidence must fall within bounds

Provenance: Every Topic Traces Back to the Transcript

Topic Index Schema

Field	Purpose
Topic ID	Stable identifier (e.g. T01)
Topic	Meaningful topic label
Start	Exact page:line reference
End	Exact page:line reference
Evidence	Supporting source references
Source chunks	Traceability to extraction stage

Example Entry

T01

Deposition ground rules and admonitions

Start 7:15

End 8:23

Evidence

7:15 · 7:24 · 8:8 · 8:18 · 8:22

A plausible topic with the wrong source location is treated as a failure.

57 topics checked · 0 provenance errors

Validation is deterministic and runs independently of the LLM — it does not rely on the model to self-report correctness.

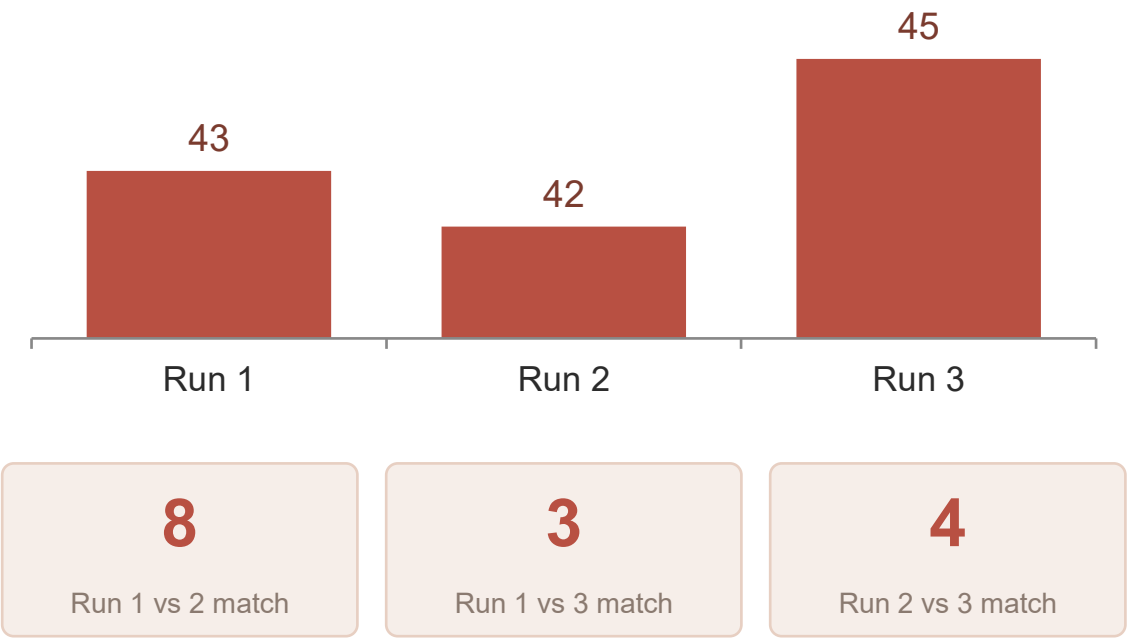
Validation & Stability

Manual Evaluation — First 20 Topics

Metric	Result
Location accuracy	20 / 20 (100%)
Topic relevance	20 / 20 (100%)
Boundary quality	17 / 20 (85%)
Coverage	20 / 20 (100%)
Redundancy	19 / 20 (95%)

Reviewed against defined criteria: does the reference point to the right location, does the label match the testimony, and is the boundary reasonable without unrelated material.

Three-Run Stability — Same Deposition, Same Pipeline



0 invalid boundary references and **0 invalid evidence** references across all three runs

Key finding: LLM segmentation varied mainly in topic granularity and boundary selection — source addressing itself remained valid in every run.

Failure Analysis, Limitations & Next Steps

Portfolio Transfers

Merged into one topic in some runs, split into two in others.

Vervent / PEAKS Loans

Granularity varied — broad topic vs. two related sub-topics.

Cancellation Rights

Topic boundaries varied by several lines across runs.

Root cause: semantic segmentation consistency, not provenance validity — every case above still had valid page:line references.

Current Limitations

- Topic segmentation can vary between runs
- Manual review covers 20 of 57 topics
- Global topic re-entry detection is partial
- Single-deposition scope so far

Application & Next Steps

- Streamlit app: search, evidence, transcript navigation
- Next: systematic boundary scoring
- Next: stronger cross-batch re-entry detection
- Next: hierarchical parent / child topics